COMPSCI 2XC3 Final Lab Report

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The A* Algorithm is an algorithm used find the shortest path from a start node to a target node in a directed weighted graph. It's algorithm is practically the same as Dijkstra's, the only difference is how it determines priority (the key value) in the priority queue.

In A*, the key value used in the priority queue is equal to the total weight of the edges along the path traversed (like Dijkstra's) plus some value coming from a heuristic function which is a lower bound to the weight of the rest of the path to the target.

A* does this to address a problem that Dijkstra has where it explores nodes that are obviously not on the shortest path. For example if I were to calculate the shortest route from Toronto to Hamilton with Dijktra's it'll probablys venture out to Scarborough before finding Hamilton which is obviously wrong. If I were to use A* though, I could use a heurisitc function that calculates the straight line distance to Hamilton. This heurisitc function would make A* prioritize exploring western locations (relative to Toronto) aside from exploring Scarborough. This would result in the shortest path being found much quicker.

You could empirically test the speed difference between A* to Dijkstra's by generating a varying number of points in 2D space, with random edges going between the two and running the algorithm on random starting and target nodes, comparing the execution times. It is important to note that for any node, it's heuristic value (straight line distance in this case) must be a lower bound to the sum of the weights of the path to the target node, otherwise A* could break and return a path that isn't the shortest.

If you generated an arbitrary heuristic function, A* could return the wrong result (same reason mentioned prior) and could also take longer then Dijktra's.

Applications where A* should be used is when there exists a heurstic function that is quick to compute, is always a lower bound to the weight of the shortest path from the inputted node to the target node, and the function decreases as you move to nodes closer to the target. This is why A* is so good for shortest route calculations because straight line distances are always the lower bound to the shortest route, straight line distances are easy to calculate ($O(1)$ via pythagorean theorem if you have cartesian coordinates) and it decreases as you get closer to the target node.

Part 3

Data and Graph Construction

We built a weighted graph from the London Tube data. Each station was treated as a node, and each connection between two stations was treated as an edge. Using `london_connections.csv`, we added an edge whenever two stations were connected in the dataset. The weight of each edge was based on the geographical distance between the two stations, which we computed using the latitude and longitude values in `london_stations.csv`. In our implementation, these connections were treated as bidirectional.

This gave us a real transportation network to work with instead of a random graph. Because the London Tube has both physical layout and line transfer structure, it was a good dataset for comparing Dijkstra's algorithm and A* in a meaningful way.

Heuristic Function

For each destination station d, we built a heuristic dictionary for A*. The heuristic value of a node was the straight line distance from that node to d. This matches the suggestion in the project description to use the physical "as the crow flies" distance as the heuristic.

This is a reasonable heuristic for the London Tube graph because every station has real coordinates. A station that is closer to the destination in physical distance is often a better direction to explore. Because of this, A* gets useful guidance toward the goal, while Dijkstra does not use this extra information.

Experimental Setup

We compared Dijkstra and A* on many source destination pairs in the London Tube graph. For each pair, we recorded the runtime of both algorithms, rebuilt the shortest path, measured the path length, and counted the minimum number of line transfers on that path. We grouped the journeys into three categories:

- **Same line**: 0 transfers,
- **One transfer**: 1 transfer,
- **Several transfers**: 2 or more transfers.

Before studying the performance results, we checked that the implementation was correct. Our test suite showed that the London data loaded correctly and that A* matched Dijkstra on sampled and additional source destination pairs. After generating the full `part3_results.csv` file, we also checked the entire output. This check found 0 distance mismatches between Dijkstra and A*, and 0 invalid rows with infinite distance or empty paths. Therefore, the results below are based on correct shortest path outputs.

Results

Figure 2 shows the median runtime of both algorithms grouped by shortest path length. In general, the median runtime increases as the path length increases. However, A* stays faster than Dijkstra across almost the whole range of path lengths. This shows that the geographical heuristic is still useful even for longer journeys.

Figure 1 shows the median runtime grouped by transfer category. Same line trips are the fastest for both algorithms, trips with one transfer are slower, and trips with several transfers are the slowest. In all three categories, A* has a lower median runtime than Dijkstra.

Figure 3 shows the speedup of A* over Dijkstra, where

$$\text{speedup} = \frac{\text{Dijkstra runtime}}{\text{A* runtime}}.$$

The median speedup is about 1.3× for same line trips, about 1.8× for trips with one transfer, and about 1.75× for trips with several transfers. This means that A* is faster in all three categories, and its advantage becomes larger when the trip is less direct.

We also looked at runtime grouped by the exact number of transfers. The same pattern appeared there too: as the number of transfers increased, the runtime of both algorithms usually increased, but A* still stayed faster than Dijkstra.

Discussion

Overall, our experiments show that A* was consistently faster than Dijkstra on the London Tube graph. The gap between the two algorithms was smallest for short same line trips, where both algorithms had less work to do. These were the cases where the two algorithms were the most similar. Even in these simpler cases, A* still had an advantage.

As trips became longer or needed more transfers, the difference between the two algorithms became larger. The main reason is the geographical heuristic used by A*. Since the graph has real coordinate data, the straight line distance to the destination gives a meaningful estimate of how far a node is from the goal.

This helps A* move in the right direction, while Dijkstra explores the graph more evenly and may spend more time on nodes that do not help reach the destination quickly.

The transfe based results support this idea. Same line trips are usually simpler and have fewer possible route choices. Trips with one or several transfers are more complicated and create more possible intermediate routes. In these harder cases, the heuristic becomes more useful because it helps A* avoid wasting time on less relevant parts of the graph. This matches the speedup values in Figure 3, where A* shows a bigger advantage once the trip is no longer a direct same line journey.

The path length results show the same idea in another way. Shorter paths are usually easier because the algorithms can finish quickly. As the path length grows, the search space becomes larger and there are more choices to consider. Because of this, the median runtime of both algorithms increases. However, the increase is much larger for Dijkstra than for A*. This suggests that the heuristic becomes more useful as the trip becomes longer and less direct.

In our experiments, Dijkstra did not show strong evidence of being better than A* on typical journeys. The closest cases were short same line trips, where their runtimes were the most similar. For this dataset, A* appears to be the better practical choice when the goal is to find a path to one specific destination and a useful heuristic is available.

We included the line based part of the assignment by using the transfer count of the rebuilt shortest path. We treated 0 transfers as a same line trip, 1 transfer as a one transfer trip, and 2 or more transfers as a trip with several line changes. This helped us connect the algorithm results directly to the structure of the London Tube system instead of only describing the graph in a general way.

Limitations

One limitation of this experiment is that the edge weights were based on geographical distance instead of real travel time. Because of this, the graph does not include waiting time, different train speeds, station delays, or walking time during transfers. In the same way, the heuristic only uses straight line distance and does not directly include transfer cost. Even with these simplifications, the heuristic still gave A* a clear advantage on this dataset.

Conclusion

In conclusion, the London Tube experiments show that A* is usually the better choice when a useful heuristic is available. On this dataset, A* had lower median runtime than Dijkstra across different path lengths and across different transfer categories. The difference was smallest for simple same line trips and larger for trips with one or several transfers. These results support the idea that A* is especially useful on real transportation networks where a good lower bound heuristic can be computed easily.

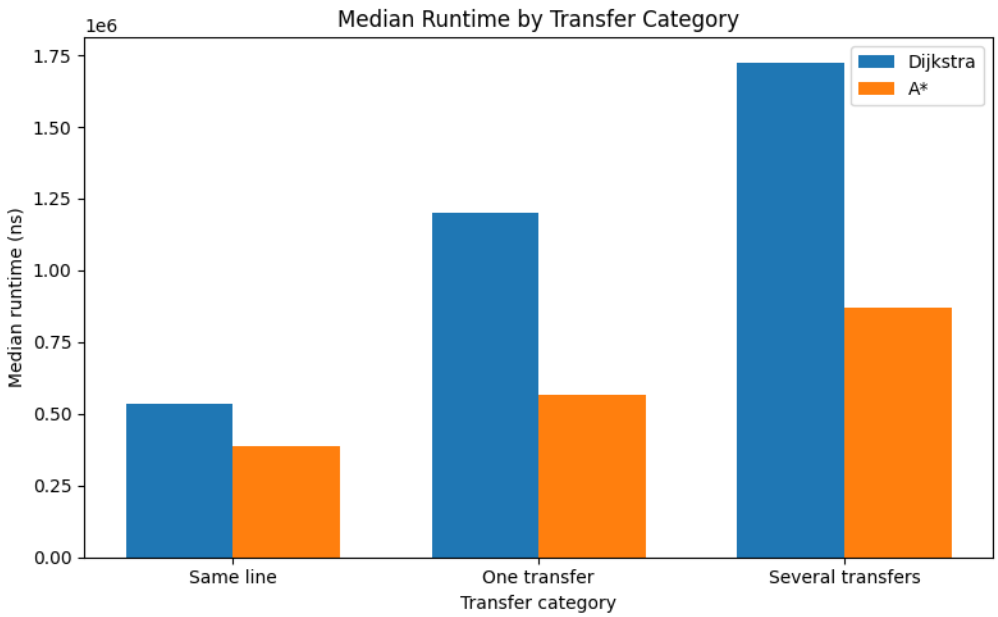

Figure 1: P3_Median runtime by transfer category. A* is faster in all three groups.

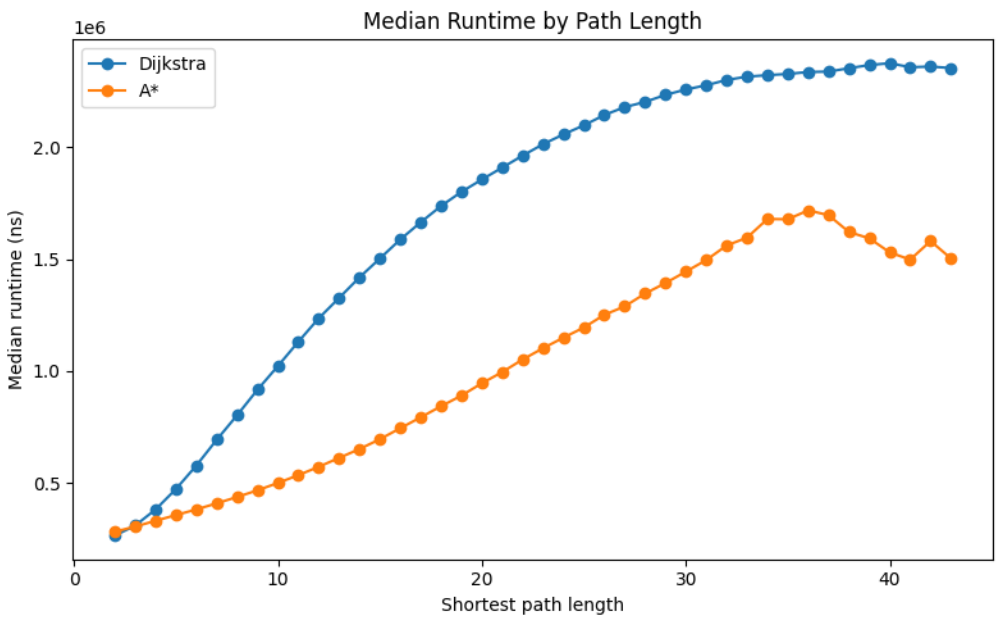

Figure 2: P3_Median runtime by path length. A* remains faster across almost all path lengths.

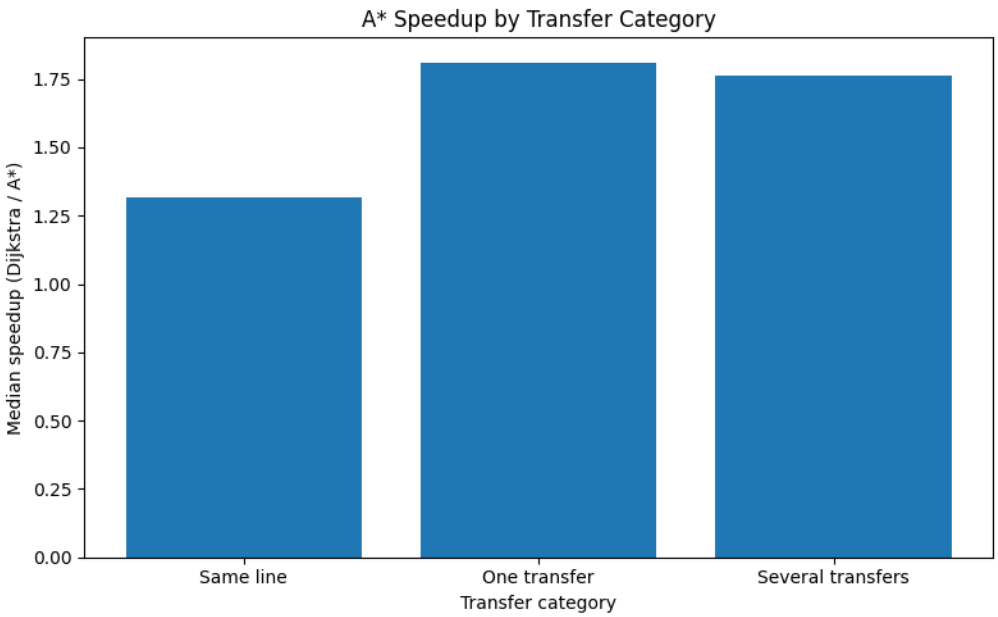

Figure 3: P3_A* speedup by transfer category.

Appendix

Additional Figures for Part 3

The following figures provide extra detail for the Part 3 analysis. They are included in the appendix because they are useful for completeness, but they are less central to the main conclusions than Figures 1, 2, and 3.

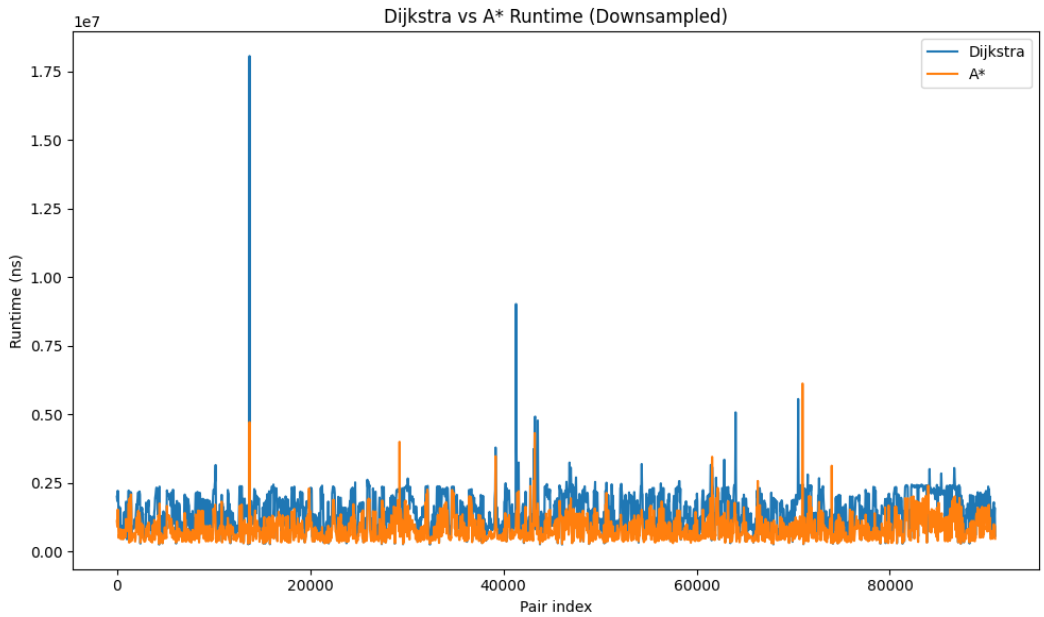

Figure 4: P3_Downsampled runtime comparison of Dijkstra and A*.

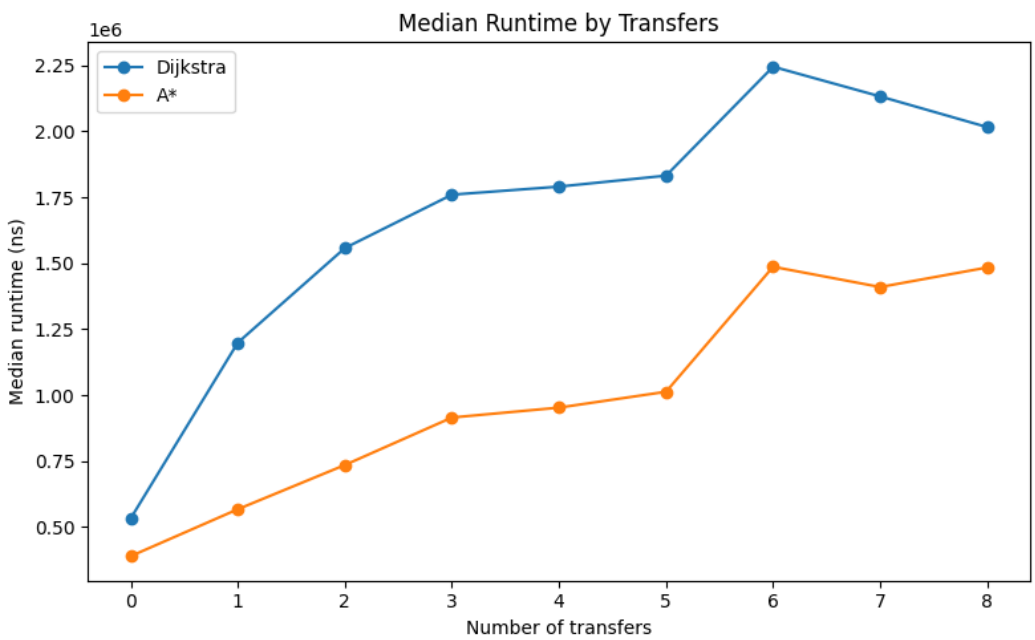

Figure 5: P3_Median runtime by exact number of transfers.

Code Navigation for Part 3

The code for Part 3 is organized as follows:

- `final_project_part3.py`: loads the London Tube data, builds the weighted graph, creates the heuristic function, and runs the main Dijkstra vs A* experiment.
- `final_project_part3_test.py`: contains correctness tests for data loading and shortest-path comparisons between Dijkstra and A*.
- `final_project_part3_resultVerification.py`: checks the generated `part3_results.csv` file for distance mismatches and invalid rows.
- `final_project_part3_test_visualization.py`: reads the experiment output and generates the figures used in the report.
- `part3_results.csv`: stores the experiment results used for the analysis and plots.